**Set up a basic IDS like Snort and test its effectiveness in detecting different types of attacks**

**Snort** is a **widely used open-source Intrusion Detection System (IDS)** and **Intrusion Prevention System (IPS)**. It monitors network traffic in real time and detects suspicious or malicious activity using predefined rules.

Snort is a **network security tool** that:

- Watches incoming and outgoing traffic
- Analyzes packets
- Alerts when it finds attacks or unusual behavior

**How Snort Works**

Snort processes network data in three main stages:

1. **Packet Capture**
   - Captures live network traffic from interfaces (like eth0)
2. **Preprocessing**
   - Cleans and organizes packets
   - Detects anomalies (fragmentation, encoding tricks)
3. **Detection Engine**
   - Compares traffic against **rules/signatures**
   - Generates alerts if a match is found

## STEP-BY-STEP PROCEDURE

- Install Snort
  - sudo apt update
  - sudo apt install snort -y
  - During installation: Enter your network interface (e.g., `eth0` or `wlan0`)
- Check installation
  - snort -V
- Configure Snort
  - Edit configuration file: sudo nano /etc/snort/snort.conf
- Set your network range:
  - ipvar HOME_NET 192.168.1.0/24
  - Replace with your actual IP range: ifconfig
- Add Custom Rules
  - Open local rules file: sudo nano /etc/snort/rules/local.rules
  - Add these rules:
  - **ICMP (Ping) Detection**
    - `alert icmp any any -> $HOME_NET any (msg:"Ping detected"; sid:1000001;)`
  - **Port Scan Detection**
    - `alert tcp any any -> $HOME_NET any (flags:S; msg:"Port scan detected"; sid:1000002;)`
  - **Brute Force (SSH) Detection**
    - `alert tcp any any -> $HOME_NET 22 (msg:"SSH brute force attempt"; sid:1000003;)`

    - Save and exit.
- Run Snort in IDS Mode
    - sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
- Simulate Attacks Use another system (or same machine carefully):
- **Test 1: Ping Attack**
    - `ping <target_ip>`
    - 📌 Expected Output: `Ping detected`
- **Test 2: Port Scan using Nmap**
    - `nmap -sS <target_ip>`
    - 📌 Expected Output: `Port scan detected`
- **Test 3: SSH Brute Force (Simulation)**
    - `hydra -l root -P passwords.txt ssh://<target_ip>`
    - 📌 Expected Output: `SSH brute force attempt`
- Analyze Logs
    - Snort logs are stored in: /var/log/snort/
    - View alerts: sudo cat /var/log/snort/alert